Brahadeesh Sankarnarayanan
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RESEARCH INTERESTS

I work in extremal combinatorics and graph theory. My current interests are in spectral extremal graph theory, intersection graphs of paths, intersecting set systems, graph convexity, majority coloring games, and list colorings.

Current Projects

Majority coloring game on trees
With Yash Chawda (IIT Jodhpur), Saraswati Girish Nanoti (IISc) and Eshwar Srinivasan (Amrita Vishwa Vidyapeetham, Coimbatore). Investigating optimal strategies and computational complexity for the two-player majority coloring game, particularly on binary and ternary trees.

$\triangle$-convexity in graphs
With Vishnu Kumar (IIT Jodhpur). Investigating convexity structures in graphs, particularly computing dimensional parameters such as the exchange number and Carathéodory number for $\triangle$-convexity, and characterizing the extremal examples.

Brualdi–Hoffman-type extremal problems
With M. Rajesh Kannan (IIT Hyderabad). Investigating spectral extremal problems for F-free graphs on m edges, extending recent results for the case $F = K_3$ to $F = K_4$ and beyond.

Intersection graphs of paths
With Niranjan Balachandran (IIT Bombay), Shagnik Das (National Taiwan University) and Umesh Shankar (IISc). Investigating the path-representation number $\mathsf{prn}(G)$ of a graph G: bounds in terms of other graph parameters, algorithms for computing optimal path representations, computational complexity of the parameter for different graph classes.

Fractional intersecting families
Continuing investigation of the Balachandran–Mathew–Mishra conjecture on the maximum size of fractional θ-intersecting families over $[n]$, building on Balachandran–Das–Sankarnarayanan (2025) and Balachandran–Bhattacharya–Kher–Mathew–Sankarnarayanan (2023).

List colorings of graphs on surfaces
Investigating the gap between the choice number and the chromatic number for graphs embedded on surfaces, particularly the choice number of the toroidal square grid, building on Balachandran–Sankarnarayanan (2026) and Balachandran–Sankarnarayanan (2021).

EDUCATION

PhD (Mathematics) 2018 – 2024
Indian Institute of Technology (IIT) Bombay
Thesis: "Some problems in combinatorics: Excursions in graph colorings and extremal set theory"
Advisor: Niranjan Balachandran
CPI: 8.89/10.0

BS–MS (Dual) Degree (Major in Mathematics) 2012 – 2017
Indian Institute of Science Education and Research (IISER) Bhopal
Thesis: "Automorphic forms and Tate's thesis"
Advisor: Karam Deo Shankhadhar
CPI: 9.31/10.0

EMPLOYMENT

Assistant Professor (Grade II) 09/2025 – present
Department of Mathematics, IIT Jodhpur

Research Associate 05/2025 – 09/2025

Department of Mathematics, IIT Hyderabad
Project: "Spectral Theory of Signed Graphs"
Principal Investigator: M. Rajesh Kannan

Institute Post-Doctoral Fellow 08/2024 – 02/2025
Department of Mathematics, IIT Bombay
Mentor: Niranjan Balachandran

Project Assistant 09/2017 – 11/2017
Department of Mathematics, IISER Bhopal
Project: "Self-dual representations and signs"
Principal Investigator: Kumar Balasubramanian

PUBLICATIONS

Refereed Journal Articles

[J9] 5-*list coloring toroidal* 6-*regular triangulations in linear time* (with N. Balachandran). Discrete Applied Mathematics **386** (2026), 75–103, doi:10.1016/j.dam.2026.01.032, arXiv:2106.01634, MR5028701, Zbl 1587.05041.

[J8] *Bounded fractional intersecting families are linear in size* (with N. Balachandran and S. Das). The Electronic Journal of Combinatorics **32** (2025), no. 3, #P3.34, 10 pp., doi:10.37236/12900, arXiv:2402.14981, MR4949605, Zbl 1572.05240.

[J7] *Low-rank matrices, tournaments, and symmetric designs* (with N. Balachandran). Linear Algebra and its Applications **694** (2024), 136–147, doi:10.1016/j.laa.2024.04.006, arXiv:2401.14015, MR4736676, Zbl 1539.05071.

[J6] *Almost full rank matrices arising from transitive tournaments* (with N. Balachandran and S. Bhattacharya). Linear and Multilinear Algebra **72** (2024), no. 2, 153–161, doi:10.1080/03081087.2022.2158168, MR4703565, Zbl 1533.05147.

[J5] *On hierarchically closed fractional intersecting families* (with N. Balachandran, S. Bhattacharya, K. V. Kher and R. Mathew). The Electronic Journal of Combinatorics **30** (2023), no. 4, #P4.37, 17 pp., doi:10.37236/11651, arXiv:2211.02540, MR4672585, Zbl 1533.05268.

[J4] *An ensemble of high-rank matrices arising from tournaments* (with N. Balachandran and S. Bhattacharya). Linear Algebra and its Applications **658** (2023), 310–318, doi:10.1016/j.laa.2022.11.004, arXiv:2108.10871, MR4513895, Zbl 1506.15034.

[J3] *Note on* 4-*coloring* 6-*regular triangulations on the torus.* Annals of Combinatorics **26** (2022), no. 3, 559–569, doi:10.1007/s00026-022-00573-8, arXiv:2106.01037, MR4473892, Zbl 1497.05098.

[J2] *The choice number versus the chromatic number for graphs embeddable on orientable surfaces* (with N. Balachandran). The Electronic Journal of Combinatorics **28** (2021), no. 4, #P4.50, 20 pp., doi:10.37236/10263, arXiv:2102.06993, MR4394670, Zbl 1486.05101.

[J1] *Self-dual representations of* $\mathrm{SL}(2, F)$*: an approach using the Iwahori–Hecke algebra* (with K. Balasubramanian and K. S. Senthil Raani). Communications in Algebra **47** (2019), no. 10, 4210–4215, doi:10.1080/00927872.2019.1581211, MR3975999, Zbl 1432.20031.

Conference Proceedings

[C2] *On the majority game chromatic number of forests and other graphs* (with Y. Chawda and S. G. Nanoti). Accepted at FSTTCS 2026. arXiv:2609.23803.

[C1] 5-*list coloring toroidal* 6-*regular triangulations in linear time* (with N. Balachandran). Extended abstract. CALDAM 2023. In: A. Bagchi and R. Muthu (eds.) *Algorithms and Discrete Applied Mathematics*, Lecture Notes in Computer Science, vol. 13947, pp. 134–146, Springer, Cham (2023), doi:10.1007/978-3-031-25211-2_10, MR4573050, Zbl 07728658.
The full version appeared as [J9] above.

Preprints

[P2] *Structural and computational aspects of majority coloring games* (with Y. Chawda, S. G. Nanoti and E. Srinivasan). 2026, arXiv:2609.37796.

[P1] *Tight bounds on the Carathéodory and exchange numbers in $\triangle$-convexity* (with V. Kumar). 2026, arXiv:2609.34441.

Manuscripts in Preparation

[M1] *An extremal problem for vertex intersection graphs of paths* (with N. Balachandran, S. Das and U. Shankar). 2025.

RESEARCH SUPERVISION

PhD Students

Piyush Joshi enrolled Autumn 2026–27
Department of Mathematics, IIT Jodhpur

Yash Chawda enrolled Spring 2025–26
Department of Mathematics, IIT Jodhpur

Vishnu Kumar enrolled Autumn 2025–26
Department of Mathematics, IIT Jodhpur

Executive MTech (Data and Computational Sciences) Projects

Rayipalli Prakash 2025–26
School of Artificial Intelligence and Data Science, IIT Jodhpur
Project: "GNN-Based Automatic Calorie Identification from Food Images"

Md Parvez Rahaman 2025–26
School of Artificial Intelligence and Data Science, IIT Jodhpur
Project: "AI-Based Text Summarization and Tone Rewriting System Using BART and T5"

Summer Interns: SURAJ Program

Vaibhav Suvagiya Summer 2026
Sardar Vallabhbhai National Institute of Technology, Surat
Project: "RAMO: A Verified MaxSAT Repair Oracle for Tabu Search on Ramsey 2-Colouring, A De-Confounded Negative Result"

B. James Mejulan Summer 2026
Sardar Vallabhbhai National Institute of Technology, Surat
Project: "Generators and Relations for Discrete Groups"

Other Interns

Sai Akshar Tadimeti Summer 2026
IIT Jodhpur Reading Project: Discrete mathematics, from Rosen's *Discrete Mathematics and Its Applications*

FELLOWSHIPS, AWARDS & HONORS

Institute Post-Doctoral Fellowship 08/2024 – 02/2025
IIT Bombay

National Board for Higher Mathematics (NBHM) Research Award 2018 – 2023
Department of Atomic Energy, Government of India

Joint CSIR–UGC National Eligibility Test for Junior Research Fellowship 2017
CSIR – Human Resource Development Group; All-India Rank 20 in Mathematical Sciences

Director's Gold Medal 2017

IISER Bhopal
Awarded for outstanding all-round achievement and leadership in the BS–MS graduating class

Rashtrapati Bhavan In-Residence Program for Scholars from IITs, IISERs, and IISc 2016
Invited by the President of India to represent IISER Bhopal

Science Academies' Summer Research Fellowship 2015 & 2016
Indian Academy of Sciences, Indian National Science Academy, and National Academy of Sciences, India

World Science Conference, Israel 2015
Hebrew University of Jerusalem; member of a 15-person Indian contingent

INSPIRE Scholarship for Higher Education 2012 – 2017
Department of Science & Technology, Government of India

RESEARCH GRANTS

Research Initiation Grant (RIG) 2026 – 2027
IIT Jodhpur
Amount: Rs. 4,00,000
Role: Principal Investigator

INVITED TALKS AND PARTICIPATION

Technical Talks

Teaching logical statements at the undergraduate level: Why one common definition gets in the way 2026
Conference on the Practice of Teaching Undergraduate Mathematics, Krea University, Sri City

Intersection Graphs and Path Representations 2026
15th Prof. E. Sampathkumar Memorial Endowment Lecture, 22nd Annual Conference of the ADMA & International Conference on Discrete Mathematics, IIT Madras

Tight bounds on the Carathéodory and exchange numbers for $\triangle$-convexity spaces 2026
Annual Meru Combinatorics Conference, IIT Bhilai

Competency-based Learning in Mathematics 2026
Session at the STEM Teacher Training Program, organized (in association with Jodhpur Zila Parishad and Shiksha Vibhag) by the Centre for Education Technology, IIT Jodhpur

Nonstandard analysis: a gentle introduction 2025
Discussion Meeting on Number Theory and Combinatorics, Evolve Back, Coorg

Bisection-closed families, tournaments, and symmetric designs 2025
Geometry, Probability, and Algorithms, International Centre for Theoretical Sciences, Bengaluru

Sunflowers and symmetric designs 2025
Research Seminar, Department of Mathematics, IIT Jodhpur

Sunflowers and symmetric designs 2024
Research Associate Seminar, Theoretical Statistics and Mathematics Unit, Indian Statistical Institute, Bangalore Centre

Some recent work on fractional intersecting families 2024
Annual Meru Combinatorics Conference, Graphic Era Hill University, Bhimtal Campus

Sunflowers, tournaments, and symmetric designs 2024
Institute Post-Doctoral Fellowship Seminar, Department of Mathematics, IIT Bombay

Some recent work on fractional intersecting families 2024
CSE Research Symposium, IIT Palakkad

Popular and Expository Talks

What Is Combinatorial Thinking? 2026
Summer School for Women in Mathematics and Statistics, ICTS Bengaluru

A Combinatorial Gem of Maryam Mirzakhani: List Colorings of Planar Graphs 2026

One Day Symposium Celebrating Women in Mathematics, IIT Jodhpur

Counting the Infinite 2025
National Mathematics Day Guest Lecture, Government College Makroniya Bujurg, Sagar, Madhya Pradesh (online)

A Glimpse of Infinity 2020
National Mathematics Day Webinar, Raja Bhoj Govt. College Katangi, Balaghat, Madhya Pradesh (online)

Research Discussion Groups

Leader, Majority coloring game on trees 2026
MATRIX-MFO Tandem Workshop: Combinatorial Interchange, MATRIX, Creswick, Australia

Participant, List packing conjecture 2026
MATRIX-MFO Tandem Workshop: Combinatorial Interchange, MATRIX, Creswick, Australia

Leader, Extremal problems on fractional intersecting families 2026
Conference on Frontiers in Combinatorics – Topology, Algebra and Data-Driven Insights, BITS Pilani, Pilani Campus

EVENTS ORGANIZED

Screening of *Secrets of the Surface: The Mathematical Vision of Maryam Mirzakhani* 2026
Documentary film by George Csicsery; screening organized under the Mathematics Society as a follow-up to the One Day Symposium Celebrating Women in Mathematics
Department of Mathematics, IIT Jodhpur

Career Opportunities Beyond Academia 2026
One-day interaction session with industry mathematicians
Department of Mathematics, IIT Jodhpur

An Introductory Workshop on the Lean Theorem Prover 2026
Three-day workshop led by T. V. H. Prathamesh (Krea University); co-organized with D. Kaur and K. Goel
Department of Mathematics, IIT Jodhpur

TEACHING EXPERIENCE

Instructor: IIT Jodhpur

MAL7060 Statistics for Data Science — Autumn 2026–27

MAL6150 Probability and Statistics — Autumn 2026–27

MAL3040 Statistics — Autumn 2026–27

MAL2040 Design and Analysis of Experiments — Spring 2025–26
49 students; overall instructor rating 3.70/5.0 (mean), median 4 across all dimensions

Teaching Assistant: IIT Hyderabad

MA5020 Functional Analysis — Autumn 2025–26

Teaching Assistant: IIT Bombay

SI 419 Combinatorics — Autumn 2024–25, Autumn 2021–22
SI 416 Optimization — Spring 2021–22
MA 205 Complex Analysis — Autumn 2021–22, Autumn 2020–21, Autumn 2019–20
MA 207 Differential Equations II — Autumn 2020–21, Autumn 2019–20
MA 108 Differential Equations — Summer 2020–21, Spring 2019–20
MA 106 Linear Algebra — Spring 2020–21, Spring 2019–20
MA 111 Calculus II — Spring 2020–21

Other Teaching and Mentoring

Summer School for Women in Mathematics and Statistics (Resource Person) 2026
International Centre for Theoretical Sciences, Bengaluru

Masterminds Program (Trainer) 2023–24, 2024–25
UAE IMO selection camp, Ministry of Education, UAE
Topics: combinatorics and number theory

Self-Assessment Team (Member) 2023
Mathematics Training and Talent Search Programme (MTTS)

Online Foundation Course in Mathematics (Mentor) 2023
Mathematics Training and Talent Search Programme (MTTS)

Online Short Course in Linear Algebra: Application View (Mentor) 2023
Mathematics Training and Talent Search Programme (MTTS)

Study-Help Room (Co-organizer) 2023
Department of Mathematics, IIT Bombay
Weekly drop-in math help for BS and MSc students

TA Skill Enhancement & Training (Master TA) 2022
Parimal and Pramod Chaudhari Centre for Learning and Teaching, IIT Bombay

A Basic Course in Number Theory (TA) 2020
NPTEL Online Certification course

PROFESSIONAL SERVICE

Memberships

Life Member, Academy of Discrete Mathematics and Applications	2026 – present
Associate Fellow, The Institute of Combinatorics and its Applications	2026 – present
Life Member, The Indian Mathematical Society	2026 – present
Life Member, Mathematics Teachers' Association (India)	2026 – present
Affiliate Member, American Mathematical Society	2026 – present

Reviewing

Hacettepe Journal of Mathematics and Statistics	2026 – present
Theoretical Computer Science	2026 – present
Indian Journal of Pure and Applied Mathematics	2025 – present
Acta Informatica	2025 – present
Proceedings – Mathematical Sciences (Indian Academy of Sciences)	2024 – present
Mathematical Reviews / MathSciNet	2023 – present
zbMATH Open	2022 – present
Conference on Algorithms and Discrete Applied Mathematics (CALDAM)	2020 – present

Session Chair

Keynote session 2026
K. Kant (ABV-IIITM Gwalior): *Projection methods for Volterra and Fredholm integral equations with mixed weakly singular kernels*
A. Jha (IIT Gandhinagar): *Domain decomposition methods for implicit solvation models*
Workshop on Advanced Methods in Integral Equations and Their Applications, IIT Jodhpur

5th Prof. R. Balakrishnan Endowment Lecture 2026
R. Venkatesh (IISc): *How to obtain the Tutte polynomials of graphs from representation-theoretic methods?*
22nd Annual Conference of the ADMA & International Conference on Discrete Mathematics, IIT Madras

INSTITUTIONAL SERVICE

IIT Jodhpur

Member, Department Postgraduate Committee (DPGC) 2026 – present
Department of Mathematics

Faculty Advisor, MTech (Data and Computational Sciences) 2026 – present
Batch of 2026–2028, Department of Mathematics

Member, Department-level committee for projection of new Exec. BS program 2026 – present
Executive BS in Mathematics and Computing, Department of Mathematics

Member, Department-level committee for projection of new BS program 2025 – present
BS in Mathematics and Computing, Department of Mathematics

Faculty Advisor, Mathematics Society 2025 – 2026
Board of Departmental Societies

IIT Bombay

Coordinator, Students' Seminar 01/2023 – 05/2023
Department of Mathematics

IISER Bhopal

President, Students Activity Council 2016 – 2017

Coordinator, Mathematics Club 2013 – 2015

CONFERENCES, WORKSHOPS & EXTERNAL COURSES ATTENDED

MATRIX–MFO Tandem Workshop: Combinatorial Interchange 2026
MATRIX, Creswick, Australia (held simultaneously with MFO, Oberwolfach)

5th International Conference on Enumerative Combinatorics and Applications 2026
University of Haifa, Israel (online)

Conference on Frontiers in Combinatorics – Topology, Algebra and Data-Driven Insights 2026
BITS Pilani, Pilani Campus

Indian School on Logic and Applications (ISLA) 2026
Krea University, Sri City

Annual Meru Combinatorics Conference 2026
IIT Bhilai

LeanLang for Verified Autonomy Hackathon 2026
Emergence Research Lab, IISc Bangalore

Indo-Polish Pre-Conference School on Algorithms and Combinatorics 2026
IIT Dharwad

Discussion Meeting on Number Theory and Combinatorics 2025
Evolve Back, Coorg

Topics in Geometric Combinatorics (NCM Workshop) 2025
Chennai Mathematical Institute

Annual Meru Combinatorics Conference 2025
BITS Pilani, K. K. Birla Goa Campus

Geometry, Probability, and Algorithms 2025
International Centre for Theoretical Sciences, Bengaluru

HDXs and Codes 2025
International Centre for Theoretical Sciences, Bengaluru

Lean for the Curious Mathematician 2025
International Centre for Theoretical Sciences, Bengaluru

Sparse graphs: treewidth, planarity, bounded expansion (GIAN) 2025
IIT Dharwad

Combinatorial Methods in Enumerative Algebra 2024
International Centre for Theoretical Sciences, Bengaluru

Discrete Integrable Systems 2024
International Centre for Theoretical Sciences, Bengaluru

Annual Meru Combinatorics Conference 2024
Graphic Era Hill University, Bhimtal Campus

A First Second Course in the Probabilistic Method 2024
IIT Palakkad

Annual Meru Combinatorics Conference 2023
Pondicherry University

Conference on Algorithms and Discrete Applied Mathematics (CALDAM) 2023
Dhirubhai Ambani Institute of Information and Communication Technology, Gandhinagar

Indo-Dutch Pre-Conference School on Algorithms and Combinatorics 2023
Dhirubhai Ambani Institute of Information and Communication Technology, Gandhinagar